LED Cochlear Implant Thermal Modeling: Summer 2026 Progress

ECE 3901/3902/3903/4699 Special Problems

Shivang Ahlawat, Yash Varakala, Anthony He, Ishan Mahajan, Ajay Murugappan

Instructors: Dr. Paul Voss & Dr. Ashutosh Srivastava

Introduction

Goal: Improve the COMSOL Multiphysics simulation of an LED cochlear implant

Simulation work done at the Institute Lafayette makerspace

Thermal safety matters: tissue heating must stay under the 3 °C limit

Summer 2026 built directly on the Spring 2026 model

Previous Work (Spring 2026)

- Added a gold strip beneath the LED array.
  - exterior gold heat source (0.000941 W) pushed temperatures higher
  - reached ~311.5 K by 10s vs. ~310.75 K for the LED-only model.
- Fixed the geometry of the gold insert extending beyond the cochlea
  - Produced more reasonable and consistent heating trends
- Tested multiple time-step sizes (0.001s, 0.01s, 0.1s) for each model configuration
  - Results converged regardless of step size,
  - Simulations are mesh/time-step independent

Summer 2026 Objectives

- Flipping LED duty cycles.
  - See how heat flow patterns were changed compared to the previous fixed power sources.
- Test the simulation for an extended duration.
  - Observe what happens as it gets closer and closer to thermal equilibrium.
- Most of our work went extensively into flipping the LED duty cycles.
- We tried testing the simulation for longer periods of time but did not produce anything significant.

The Fix: Making Heat Sources Pulse

- Our proposed solution:
  - Create two parameters, one for duty cycle fraction and the other for the pulse period
  - Set pulse = mod(t,T_pulse) < dc*T_pulse
  - When pulse = 1 (LED is on), multiply by each individual LEDs' power value
  - To flip this cycle, we would just set the duty cycle fraction equal to 1 minus itself.
- Solution still needs to be tested

Findings & Limitations

- Transparent result: we did not reach our intended progress on flipping the duty cycle
- The proposed solution accounts for pulsating power, but still needs full implementation in the simulation before conclusions can be drawn
- Fixed max step size is a risk: a step sized for dc = 0.4 may skip entire transitions at much shorter duty cycles
- Skewed duty cycles (very short or very long) may make the pulse variable inaccurate
- The extended-duration objective went untested, the month went to the duty cycle problem

What We Learned

- COMSOL Multiphysics: parameters, variables, heat-source features, and time-dependent studies
- Transient heat transfer in biological tissue and how heat dissipates through implant materials
- Duty cycles and pulsed power: the difference between averaged power and true on/off switching
- Numerical solver behavior: time-stepping, max step constraints, and the Events interface for discontinuous loads
- Mesh density and time-step choices as reliability/accuracy trade-offs
- Bioheat safety standards, evaluating against the 3 °C limit at the hottest spot, not the average
- Research practice: question the assumptions baked into an inherited model before trusting its outputs

Future Objectives

- **Power values:** rebuild each LED source using on-state power, since current values are already averaged
- **Quick check:** compare a pulsed run against the Spring 2026 results
- **Solver settings**: use COMSOL's Events interface so switches aren't skipped
- **Symmetry testing**: flip duty cycles across the array and in time, then compare
- **Extended duration:** use a stationary study to reach the final temperature faster
- **Boundary refinement:** finish the air bubble case, refining the mesh only near the air
- **Safety check:** compare to the 3 °C limit using the hottest spot